

Toward Quantum Competitiveness in Adiabatic Quantum Computing: A Comparative Study of Variational Algorithms

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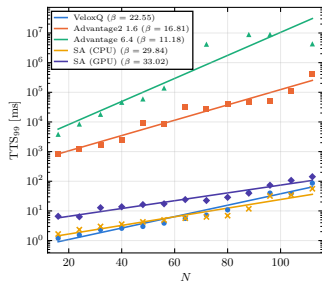
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Local Context

- Institute of Applied and Theoretical Informatics, Polish Academy of Sciences
- Quantum Systems for Informatics group
- Work on Adiabatic Quantum Computing, simulation of many-body quantum systems

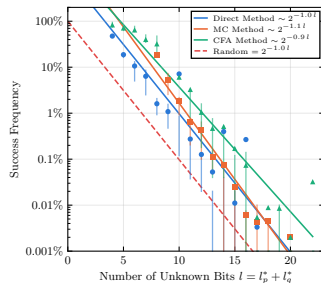


- Previous results on *exact solving* were not very promising
- Problems where a wide range of samples, not necessarily the exact solution, can be exploited



(a) Time-to-success, quantum simulation

doi:10.1103/rff3-jm5m



(b) Success rate prime factorization

arXiv:2410.14397

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Quantum Ising Models

TFIM Hamiltonian [1]

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x$$

- will come back to this later when talking about Quantum Annealing ;)

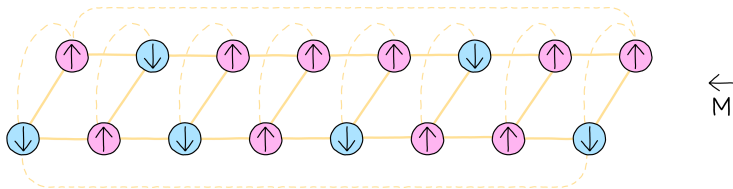


Image: PennyLane Datasets, <https://pennylane.ai/datasets/transverse-field-ising-model>

Heisenberg Hamiltonian [2]

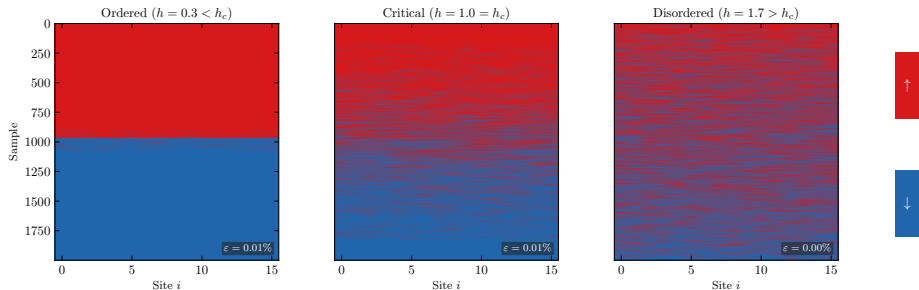
$$H = J_1 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+1} + J_2 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+2}$$

$\boldsymbol{\sigma}_i = (\sigma_i^x, \sigma_i^y, \sigma_i^z)$, so $\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j = \sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z$

- main difference to Ising: spin exchange instead of flipping, next nearest neighbor coupling
- goal in both cases: finding the **ground state** (and sampling from it)

TFIM Hamiltonian [1]

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x$$



samples from a trained model with 16 spins

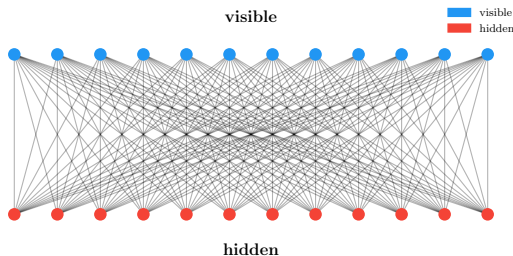
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Restricted Boltzmann Machines

- classical neural network with a layer of visible and hidden units [3, 4, 5]

RBM energy function

$$E_{\theta}(\mathbf{v}, \mathbf{h}) = \mathbf{a} \cdot \mathbf{v} - \mathbf{b} \cdot \mathbf{h} - \mathbf{h} \cdot \mathbf{W}^{\top} \cdot \mathbf{v}$$



Fully connected RBM with 12 visible and 12 hidden units

1. Sampling from the Boltzmann distribution

RBM wavefunction: hidden units traced out

$$|\Psi\rangle = \sum_{\mathbf{v}} \Psi(\mathbf{v}) |\mathbf{v}\rangle, \quad \Psi(\mathbf{v}) = \sqrt{\sum_{\mathbf{h}} e^{-E_{\theta}(\mathbf{v}, \mathbf{h})}}$$

$$\Rightarrow \Psi(\mathbf{v}) = e^{-\mathbf{a} \cdot \mathbf{v} / 2} \prod_{j=1}^{N_h} [2 \cosh(b_j + \sum_i W_{ij} v_i)]^{1/2}$$

closed form [4]

Born-rule marginal

$$\rho(\mathbf{v}) = \frac{|\Psi(\mathbf{v})|^2}{\sum_{\mathbf{v}'} |\Psi(\mathbf{v}')|^2}$$

2. Calculating the local energies

Local energy [5]

$$E_{\text{loc}}(\mathbf{v}) = \frac{\langle \mathbf{v} | H | \Psi \rangle}{\Psi(\mathbf{v})}$$

TFIM local energy [5, 4]

$$E_{\text{loc}}(\mathbf{v}) = -J \sum_{\langle i,j \rangle} v_i v_j - h \sum_i \frac{\Psi(\bar{\mathbf{v}}_i)}{\Psi(\mathbf{v})}$$

3. Calculate the log derivatives

Log-derivatives [5]

with $\phi_j = b_j + \sum_i W_{ij} v_i$

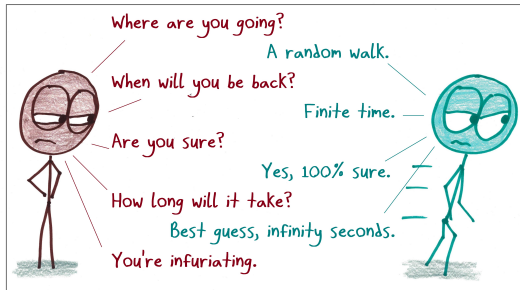
$$\frac{\partial \log \Psi(\mathbf{v})}{\partial \theta_k} = \frac{1}{2} \begin{cases} -v_i, & \theta_k = a_i \\ \tanh(\phi_j), & \theta_k = b_j \\ v_i \tanh(\phi_j), & \theta_k = W_{ij} \end{cases}$$

4. Update the parameters using GD / Stochastic Reconfiguration

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Samplers

- Classical Markov-Chain-Monte-Carlo samplers (Metropolis [6, 7]/Gibbs [8]) on GPU and FPGA
- quantum samplers: D-Wave Quantum Annealer



The key to any marriage is good communication.

Ben Orlin, *Math with Bad Drawings*,

<https://mathwithbaddrawings.com/2018/10/31/im-going-on-a-random-walk/>

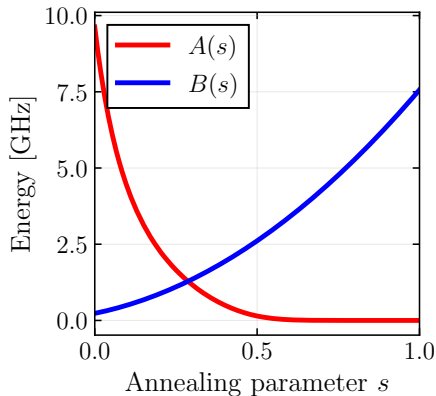
Driver Hamiltonian

$$H_D = - \sum_{i=1}^N \sigma_x^{(i)} \text{ with ground state } |+\rangle^{\otimes N}$$

Problem Hamiltonian

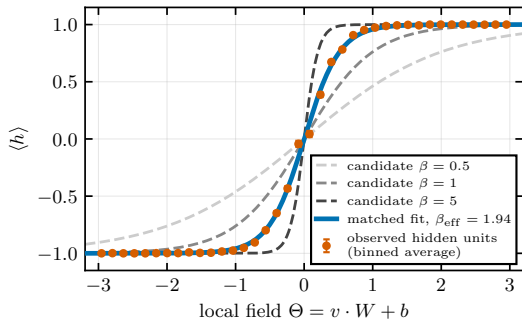
$$H_P = \sum_{i=1}^N h_i \sigma_z^{(i)} + \sum_{i=1}^N \sum_{j=1}^{i-1} J_{ij} \sigma_z^{(i)} \otimes \sigma_z^{(j)}$$

Stays in the ground state according to the adiabatic theorem [9, 10]

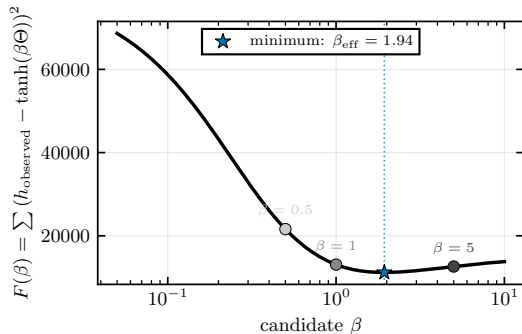


Annealing schedules $A(s)$, $B(s)$

- D-Wave sampler has unknown temperature
- CEM estimates the temperature from samples [11]



(a) Candidate fits vs. observed hidden units.



(b) Matching objective $F(\beta)$.

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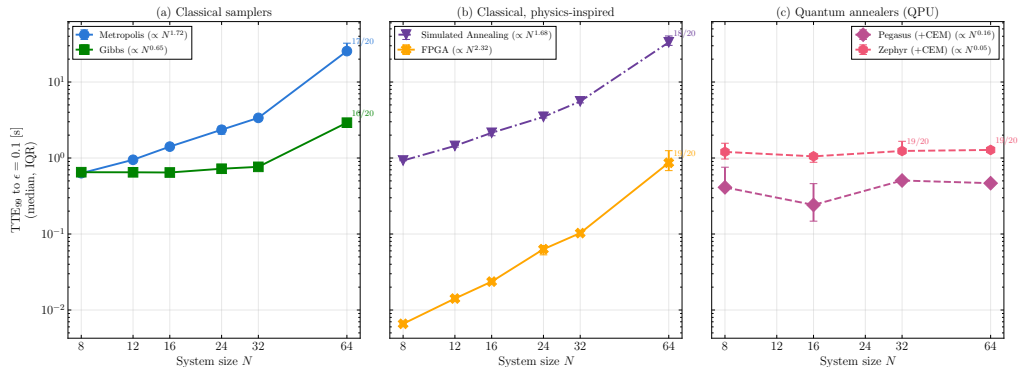
Results

- sample matched framework
- hyperparameter-matched comparison, TFIM ($h = 0.5$), N up to 64

Time-to- ϵ , 99% confidence

$$\text{TTE}_{99} = \begin{cases} T_r, & p = 1, \\ T_r \frac{\ln(1 - 0.99)}{\ln(1 - p)}, & 0 < p < 1, \\ \text{undefined (censored)}, & p = 0. \end{cases}$$

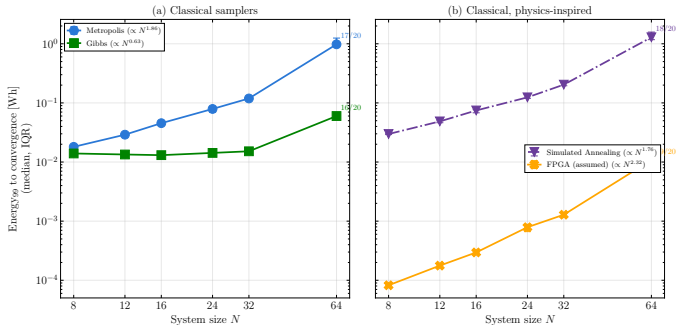
TTE₉₉ at convergence ($h=0.5$, $lr=0.08$, $reg=0.05$, $ns=200$)



Pegasus and Zephyr (+CEM) overtake FPGA by the largest system size tested

Energy consumption results

Energy₉₉ at convergence ($h=0.5$, $lr=0.08$, $reg=0.05$, $ns=200$)
D-Wave QPU omitted – no per-job energy telemetry available



FPGA is the fastest and most energy-efficient of the metered classical solvers

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https://www.pks.mpg.de/fileadmin/user_upload/MPIPKS/group_pages/SMC/quantum_magnetism_lecture_notes/lecture9.pdf. No publication date given in the source; [Accessed 29-06-2026].
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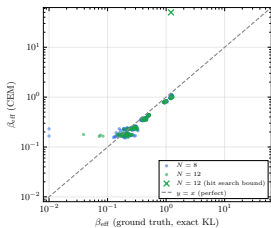
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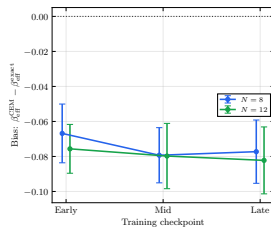
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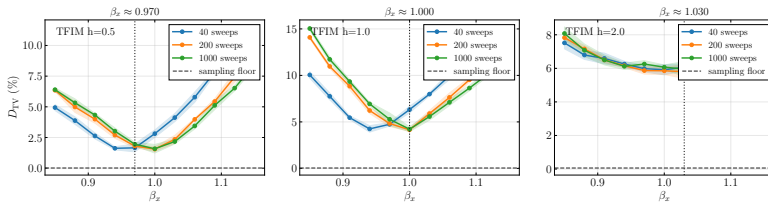
CEM & Sampling Validation



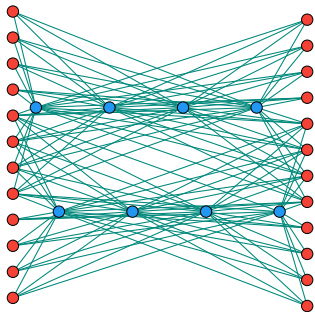
(a) CEM vs. exact ground truth



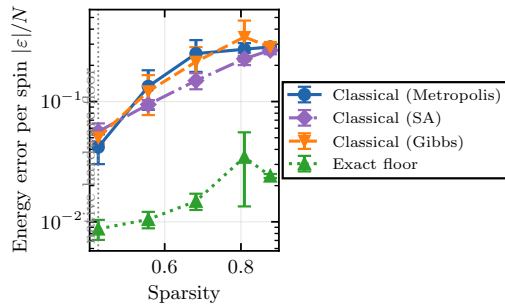
(b) Bias across training



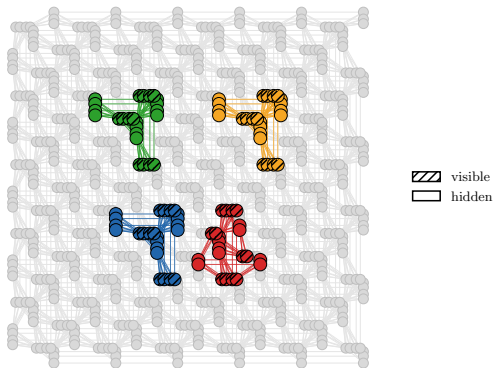
(c) D_{TV} vs. cooling sweeps



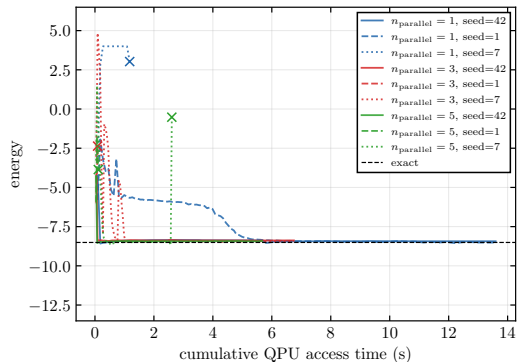
(a) Sparse RBM construction



(b) Sparsity vs. sampler error



(a) Parallel embedding layout



(b) QPU time speedup

Stoquasticity

A Hamiltonian is *stoquastic* if, in the given basis, all off-diagonal elements are real and non-positive [12].

Using $\sigma^x |\uparrow\rangle = |\downarrow\rangle$, $\sigma^y |\uparrow\rangle = i |\downarrow\rangle$, $\sigma^y |\downarrow\rangle = -i |\uparrow\rangle$, the Heisenberg exchange term acts on an antiparallel pair as

$$\left(\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y\right) |\uparrow\downarrow\rangle = |\downarrow\uparrow\rangle + (i)(-i) |\downarrow\uparrow\rangle = 2 |\downarrow\uparrow\rangle$$

Consequently, every antiparallel bond contributes

$$\langle\downarrow\uparrow| H_{ij} |\uparrow\downarrow\rangle = 2J > 0 \implies \textbf{not stoquastic}$$

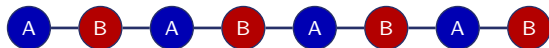
Marshall Sign Rule

- Bipartite lattice $A \cup B$: every nearest-neighbour (J_1) bond connects a site in A to a site in B
- Unitary transformation, Pauli convention [13]:

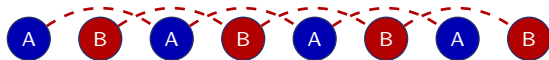
$$U = \prod_{i \in A} \sigma_i^z$$

- Ground-state wavefunction after transformation (only positive amplitudes):

$$U|\psi_0\rangle = \sum_{\sigma} |c_{\sigma}| |\sigma\rangle$$



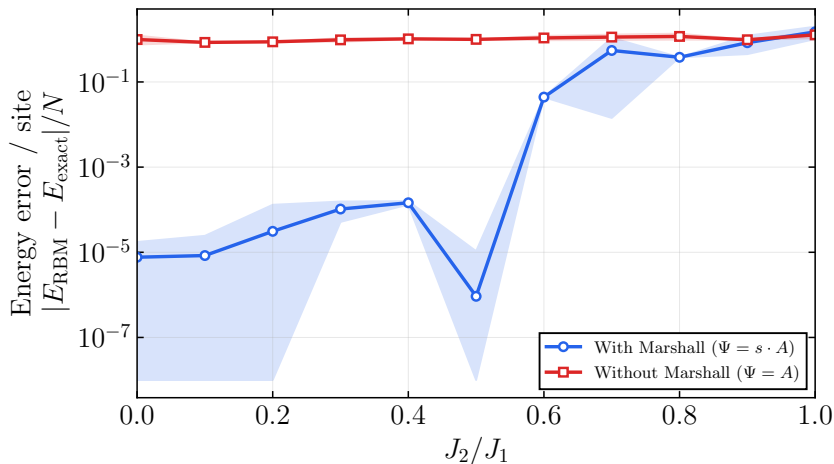
(a) J_1 : always $A-B$



(b) J_2 : same sublattice $\rightarrow$ breaks bipartiteness

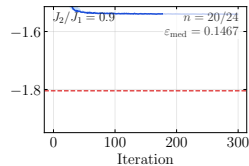
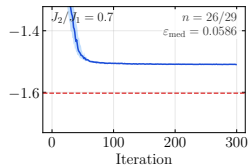
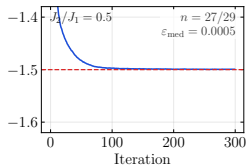
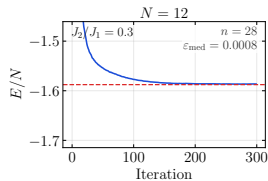
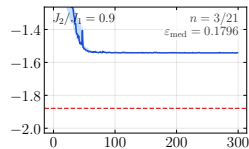
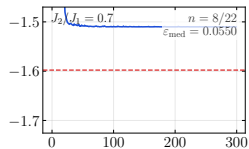
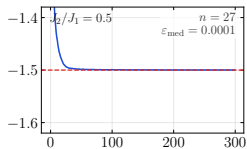
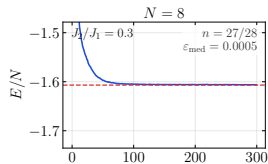
Marshall correction survives moderate frustration [14, 15]

Marshall Sign Correction



Energy error per site vs. J_2/J_1 , with/without correction

RBM Convergence Curves



E/N vs. SR iteration, frustrated Heisenberg chain